Cardiovascular attack risk prediction based on lifestyle patterns

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Abstract: Cardiovascular attacks are increasing rapidly, becoming one of the key causes of deaths in India. Early detection and prevention of these attacks have become very important. Main objective of this research is to understand and investigate that how the lifestyle choices made by individuals such as diet, physical activity, stress levels and other lifestyle related habits along with medical parameters like cholesterol and blood pressure can be analyzed to determine the risk of heart attack with the help of machine learning models. Unlike previous studies, this work compares five algorithms (Random Forest, K-Nearest Neighbour, Logistic Regression, Decision Tree, and Deep Learning) across diverse Indian demographics. Performance metrics such as Accuracy, precision, Recall and F1 score are used to evaluate machine learning algorithms. Random forest machine learning model outperformed others in identifying individuals at high risk of heart attack with an accuracy and recall rate of 97% and 99% respectively. The performance of other models was relatively mediocre and their predictive abilities reduced due to a lower recall rate. Although the Deep Learning model achieved a good precision, its slightly lower recall made it unsuitable for practical use. These results underscore the potential of ML-based methodologies in improving cardiovascular risk evaluations. However, there are challenges such as the need for model optimization and limitations in data availability. Future studies should concentrate on enhancing ML techniques, increasing dataset sizes and incorporating real-time health monitoring systems to boost predictive accuracy, early intervention measures and broader application.

Keywords— Cardiovascular attack, Heart attack risk, Cardiovascular Risk Prediction, Lifestyle patterns, Big data analytics, Machine learning, Predictive modelling, Preventive healthcare.

I. INTRODUCTION

In India, cardiovascular diseases (CVDs) continue to be a major cause of death, with 32,457 fatalities recorded in 2022 alone. The interdependent relationship between lifestyle choices and cardiovascular health has gained increasing research attention. Heart attacks account for 85 percent of these deaths, highlighting a growing concern. This study seeks to create a highly precise machine learning model to forecast the likelihood of heart attacks by examining crucial lifestyle and health indicators. Factors such as high blood pressure, cholesterol levels, smoking, diabetes, hypertension, family history, alcohol consumption, physical activity levels all contribute to the risk of a heart attack.

Recent progress in artificial intelligence (AI) and machine learning (ML) has transformed healthcare, facilitating automated, data-driven diagnostic tools. By utilizing extensive datasets, ML algorithms can uncover hidden patterns in patient data and deliver highly accurate risk evaluations. This study employs a machine learning-based method to predict heart attack risk with a 97% accuracy rate.

Despite advancements in medical diagnostics, late detection leads to delayed treatment and high mortality rates in India. There is a pressing need for an accurate, predictive, and analytical model that can evaluate an individual's risk of experiencing a heart attack,

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The main goal is to develop a machine learning model which predicts likelihood of occurrence of heart attack based on daily lifestyle patterns. In this research, a Random Forest classifier is used to analyze a comprehensive dataset of 10,000 adults in India across 26 attributes covering demographics, lifestyle factors, medical history, and cardiovascular health indicators. Techniques like data balancing and feature selection are used to improve model's scalability.

Hyperparameter tuning is used to refine the model, achieve high predictive accuracy while also providing insights to the most significant contributors which leads to heart attack risk.

The structure of the remainder of the paper is as follows: Section 2 introduces literature review. Section 3 presents overview of the basic concepts and dataset used in this research. Section 4 shows the methods and steps done , while Section 5 compares and discusses about the different machine learning models. Finally, Section 6 outlines the conclusions and suggests directions for future research.

II. LITERATURE REVIEW

There is a necessity for enhanced prediction techniques to make possible the early detection and prevention of cardiovascular diseases (CVDs) as they have become one of the major reasons of deaths in India. Several researchers have explored different machine learning and artificial intelligence driven methods in order to improve accuracy of heart attack prediction. Different algorithms such as Decision Trees, Support Vector Machine, Neural Networks, K-Nearest Neighbor(KNN) and Random Forest have been assessed by scholars based on their classification accuracy and reliability metrics in order to demonstrate the effectiveness of different ML approaches in forecasting heart disease. For superior predictive performance, advanced deep learning techniques like Convolutional Neural Network(CNN) and Recurrent Neural Network (RNN) have been proposed [1],[2].

Predictive models have been significantly improved by leveraging huge amount of patient data and the integration of concepts of big data analytics in cardiology. Research's have found that utilizing AI-driven systems like Electronic Health Records (EHRs) have effectively identified patterns and factors associated with CVD's, also enabling real time risk assessment and monitoring. Ensemble learning techniques like Gradient Boosting and Random Forest are analyzed to improve reliability of predictive models [1],[3],[4].

In heart attack prediction, feature selection plays a very important role. Studies and data analysis have identified key factors as major contributors to cardiovascular risk such as – smoking habits, cholesterol levels and high blood pressure. Hybrid approaches which integrate feature engineering with ML models have been proven effective in reducing computational complexity and improving predictive accuracy. Various studies also draw attention to the significance of real time monitoring and risk evaluation with the help of wearable devices and IoT based monitoring systems. AI powered systems monitors real time physiological data, heart rate variability and ECG signal [5],[6],[8].

Some challenges still persist in areas such as data privacy, clinical implementation and model interpretability. To build trust and acceptance within healthcare setups, researchers have emphasized on the importance of explainable AI models that provide transparent decision making [10].

III. DATASET

The dataset used in this study is publicly available and fully anonymized data published on Kaggle under open-access license. It consists of 10,000 health records from young adults in India, which detail various lifestyle, medical and cardiovascular risk factors. A total of 26 attributes are analyzed and categorized into demographics, lifestyle factors, medical background, and cardiovascular health metrics.

Demographic attributes consist of age, gender, geographic region, classification as urban or rural, and socioeconomic status (SES). The dataset captures the individual's age and gender (male or female). Geographic regions are categorized into four groups: North, South, East, and West. The urban and rural classifications denote the type of living area, while SES is divided into low, middle, or high.

Lifestyle elements include smoking habits, alcohol use, dietary type, physical activity level, screen time, and duration of sleep. Smoking habits are categorized as never, occasionally, or regularly. Alcohol consumption is similarly classified. Dietary type includes vegetarian, non-vegetarian, and vegan options. Physical activity is categorized as sedentary, moderate, or high, and screen time is recorded in hours per day. Sleep duration is measured in hours per day.

Attributes related to medical history encompass the existence of a family history of heart disease, diabetes, hypertension, cholesterol levels, body mass index

(BMI), and stress levels. The dataset notes whether an individual has a family history of heart disease, diabetes, or hypertension, using a binary format (yes/no). Cholesterol levels are quantified in milligrams per decilitre (mg/dL). BMI is documented in kilograms per square meter (kg/m²). Stress levels are classified as low, medium, or high.

Cardiovascular health indicators comprise blood pressure (systolic/diastolic mmHg), resting heart rate, ECG findings, types of chest pain, maximum heart rate during exercise, presence of exercise-induced angina, blood oxygen saturation (SpO_2%), and triglyceride levels. Blood pressure is recorded with systolic and diastolic values in mmHg. Resting heart rate is expressed in beats per minute (bpm). ECG findings indicate normal or abnormal results. Chest pain is classified into typical, atypical, or non-anginal categories. The peak heart rate attained is documented during physical activity, and the occurrence of exercise-induced angina is noted as a binary variable (yes/no). The blood oxygen saturation is recorded as a percentage and the triglyceride concentration is measured in mg per dl.

The target variable, the probability of a heart attack, signifies a binary classification that denotes whether an individual is at risk of experiencing a heart attack (yes/no). This dataset functions as the foundation for developing machine learning models intended to forecast the risk of heart disease based on the aforementioned features.

IV. METHODOLOGY

A. Research Design:

In this study, Random Forest algorithm is used to predict heart attack likelihood under a supervised machine learning approach. The ability to handle nonlinear and complex data made this model a robust choice for the research. The research follows a structured flow comprising data collection, preprocessing, model training, evaluation, and performance analysis. The predictive model is built to analyze multiple cardiovascular risk factors and provide an accurate classification of individuals at risk of heart attack.

B. Data Collection:

The dataset is taken from Kaggle which is a open access platform for reliable machine learning datasets. Our dataset contains demographic, lifestyle, and clinical risk factors associated with heart disease. The key features include smoking habits, alcohol consumption, hypertension, family history of diseases, cholesterol levels, blood pressure, sleep patterns, and screen time duration.

C. Data Preprocessing:

Several preprocessing steps included handling missing values, normalizing numerical variables, label encoding and encoding categorical features. This ensured the data was suitable for input into the machine learning model. Feature Encoding was applied where categorical variables were converted into numerical format using label encoding. To address the imbalance between heart attack cases and non-cases, the minority class was upsampled using resampling techniques to avoid any kind of bias.

D. Data Analysis Techniques:

To improve model interpretability and performance, few analytical techniques were implemented like Feature Correlation Analysis. A heatmap was generated to identify strong correlations between various risk factors and heart attack occurrences. Performance evaluation is measured by using accuracy, precision, recall and F1- score performance metrics. The Random Forest model achieved the highest performance, with a 97% accuracy and F1score, and a 99% recall, indicating its effectiveness in identifying high-risk cases.

Hyperparameter Tuning is done with the help of GridSearchCV. It is utilized to optimize the parameters of the Random Forest model, enhancing its predictive accuracy.

Some statistical insights are derived with the help of scala and sql query processing from our dataset. Some of the major factors on which heart attacks depend on are smoking habits, alcohol consumption and hypertension.

E. Tools and Technologies:

The study utilized several programming tools and libraries:

Programming Language: Python

Machine Learning Models: Random Forest Classifier

Optimization Library: GridSearchCV for hyperparameter tuning

Visualization Tools: Matplotlib and Seaborn for correlation heatmaps and model performance graphs

Data Processing: Pandas and Scikit-learn for feature engineering and model implementation.

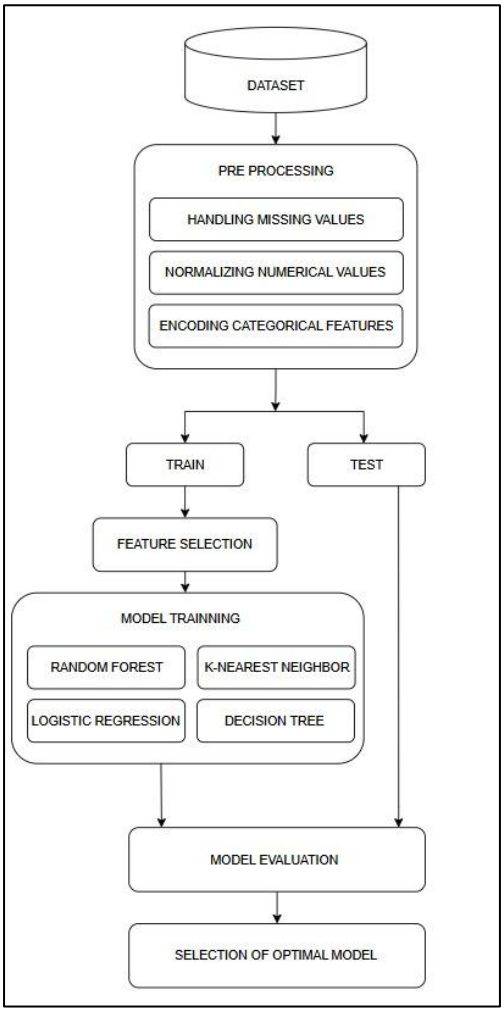

Fig 1. Flow Diagram

V. DISCUSSION

In this research, we examined five distinct machine learning algorithms—Random Forest, K-Nearest Neighbor (KNN), Logistic Regression, Decision Tree, and Deep Learning—to evaluate the risk of heart attacks among young adults based on their lifestyle choices. Each algorithm was evaluated on the basis of the basic performance indicators: accuracy, precision, recall and F1-score as shown in Table 1.

Results show Random Forest algorithm to have the highest accuracy of 97 %, the F1 score of 97 % and the recall rate of 99 %. It becomes a great choice in early detection and identification of high risk individuals. Random forest algorithm's ability to process complex data structures and reduce overfitting have been key reasons for such high performance of the algorithm.

Although, KNN had achieved a 75% accuracy and 97% recall but its performance degraded due to lower precision and F1 score, indicating effectiveness in identifying high risk individuals but also resulting in a significant number of false positives and hence making it less reliable.

The lowest performance was done by Logistic Regression with 53% accuracy, 52% precision and 56% recall, resulting in a F1 score of 54. It struggled to distinguish between different levels of risk, which makes it less useful in predicting the likelihood of a heart attack according to the results.

The Decision Tree Model was slightly better in performance than logistic regression, with a 57 percent accuracy, 52percent precision and F1 score of 59 percent. However, the 63 percent recall rate indicates that decision tree model has a better ability to detect a higher percentage of real high risk cases than the logistic regression model.

Lastly, Deep Learning model achieved 77% accuracy, 79% precision, 74% recall, and a 76% F1 score. Dropout layers are used to avoid overfitting after two hidden layers with ReLU activation. For binary classification, the output layer employs a sigmoid function. Binary cross-entropy loss and the Adam optimizer were used to train the model over 50 epochs. Further optimization is needed to improve its recall in order to detect high risk cases in a more effective manner.

Model	Accuracy	Precision	Recall	F1-Score
Random Forest	97%	95%	99%	97%
K-Nearest Neighbor	75%	67%	97%	79%
Logistic Regression	53%	52%	56%	54%
Decision Tree	57%	55%	63%	59%
Deep Learning	77%	79%	74%	76%

Table 1. Model Performance

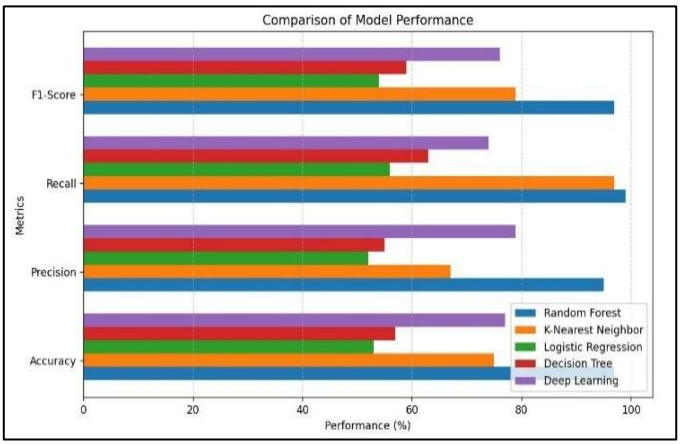

Fig 2. Comparison of Model Performance

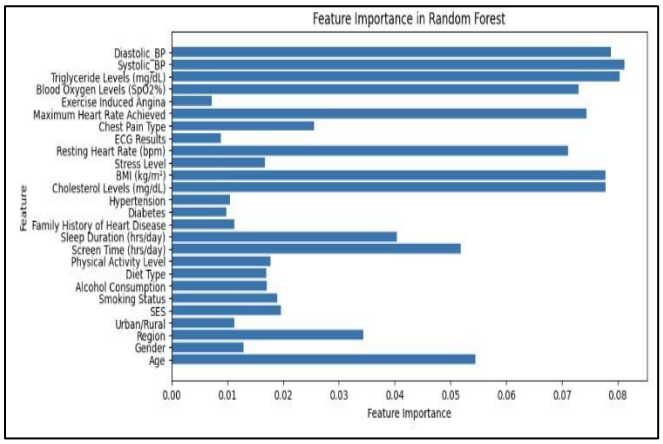

Fig 3. Feature importance

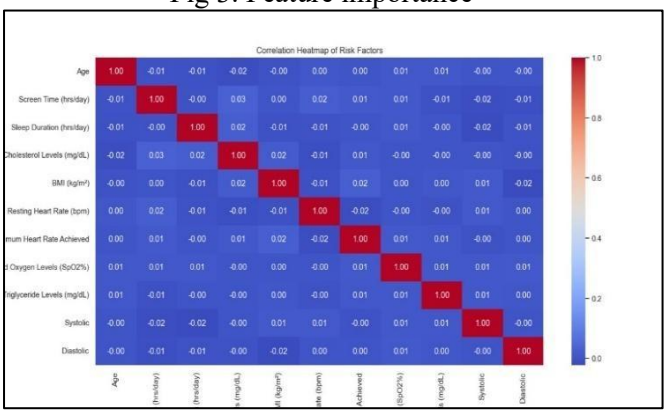

Fig 4. Heat Map

Model performance comparison shown in Figure 2 demonstrates that the Random Forest and Deep Learning models achieved the highest overall scores across accuracy, recall, and F1-score, with Random Forest performing slightly better. The Random Forest model's feature importance analysis is presented in Figure 3 which indicates that systolic and diastolic blood pressure, triglyceride levels, and resting heart rate are the most significant predictors. The correlation heatmap exhibited in Figure 4 confirms strong associations between systolic and diastolic BP, cholesterol and triglycerides, while most other features show weak correlations.

VI. KEY FINDINGS

After analyzing and visualizing the dataset , some patterns were found concerning heart attack risk factors. A significant 91percent of people who had heart attack had at least one of the following factors: smoking habit, alcohol consumption, high blood pressure, family history of heart disease or cholesterol levels exceeding 200mg/dL. This data shows that lifestyle choices are interconnected with individuals cardiovascular health.

Additionally, 43percent of people shown insufficient sleep (less than 8 hours) and excessive screen time (more than 4 hours) both of which are related with high risk of cardiovascular diseases. Prolonged exposure to screens and poor sleep patterns have resulted in stress, sedentary behaviours and metabolic problems increasing the risk of heart disease.

The key factors of heart attacks are cholesterol levels, blood pressure, and lifestyle choice. Increased cholesterol and irregular blood pressure measurements act as essential indicators, whereas lifestyle practices like smoking and alcohol consumption significantly influence cardiovascular risk.

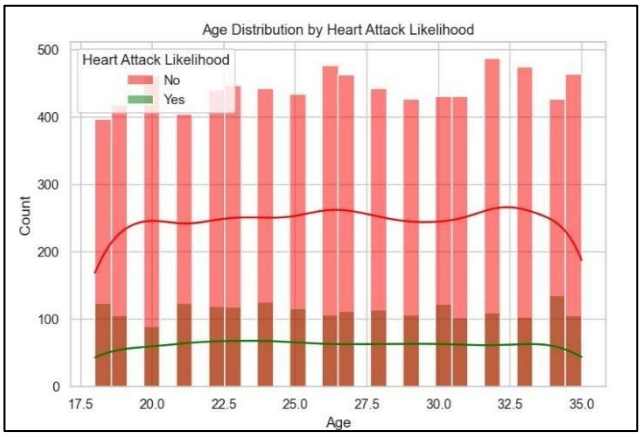

Fig 5. Age Distribution by Heart Attack Likelihood

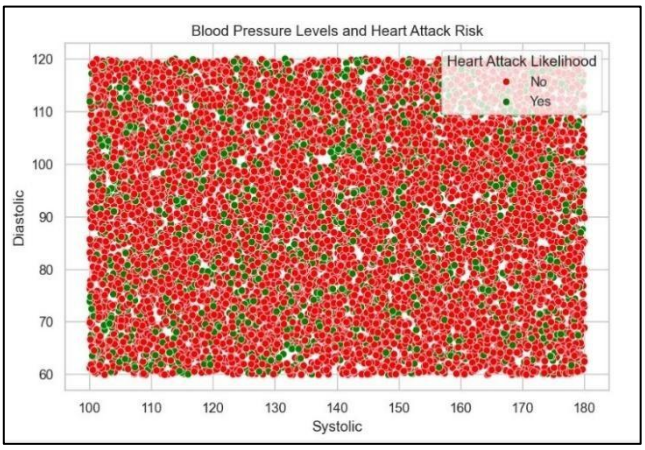

Fig 6. Blood Pressure levels and Heart Attack Risk

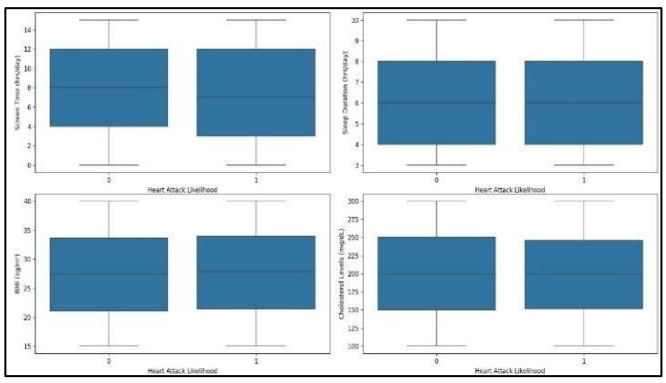

Fig 7. Box plot for numerical lifestyle factors

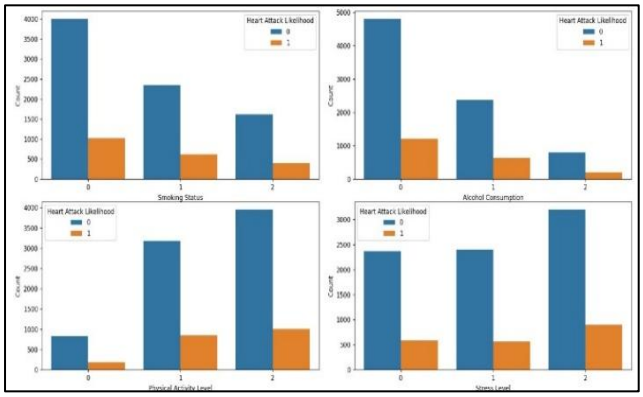

Fig 8. Count plot for categorical lifestyle factors

The age distribution chart as illustrated in Figure 5 shows that heart attacks are distributed across all age groups in adults, with a noticeable concentration between ages 25 and 32. The scatter plot in Figure 6 visualizes how individuals with elevated systolic and diastolic blood pressure are more likely to be at risk. The boxplot in Figure 7 shows that individuals with heart attack likelihood tend to have higher cholesterol levels, BMI, screen time, and lower sleep duration, indicating the subtle but impactful role of lifestyle choices. Bar charts in Figure 8 highlights that smoking, alcohol consumption, low physical activity, and high stress levels are significantly more prevalent among those at higher heart attack risk.

VII. CONCLUSION

In this research, the application of machine learning models to forecast heart attack risk in young adults by examining their lifestyle habits is explored. The study evaluated five models—Random Forest, K-Nearest Neighbor (KNN), Logistic Regression, Decision Tree, and Deep Learning using key performance metrics like Accuracy, Precision, Recall, and F1-Score. Random Forest emerged as the most accurate model with a 97% accuracy rate and a recall of 99%, making it the most dependable. In contrast, KNN, Logistic Regression, and Decision Tree exhibited moderate performance but had lower recall rates, limiting their effectiveness in identifying high-risk individuals. Although the Deep Learning model had good performance, its recall score was low, indicating a need for further improvement. These results highlight the promise of machine learning in enhancing cardiovascular risk evaluation and early intervention strategies. Nonetheless, challenges such as data imbalance and model optimization remain. Future studies should aim to broaden the dataset, include more lifestyle and clinical variables, and refine the learning models to boost recall and generalizability. Additionally, incorporating real-time health monitoring through wearable devices and electronic health records could enhance predictive accuracy and aid in proactive heart disease prevention.

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